REHOSPACE REAL ESTATE PLATFORM (RREP)

VOLUME III

IMPLEMENTATION, DEPLOYMENT & OPERATIONS GUIDE

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DOCUMENT PURPOSE

This document defines the procedures and operational standards required to implement, configure, deploy, maintain, monitor, upgrade, support, secure, and recover the RehoSpace Real Estate Platform.

Volume II defines the technical architecture of RREP.

Volume III converts that architecture into practical operational procedures.

The objective is to ensure that every standalone customer deployment can be installed and operated using a consistent, repeatable, documented process.

DOCUMENT RELATIONSHIP

The RREP documentation series is structured as follows:

Volume 0 — Product Vision & Strategy

Defines why RREP exists, its strategic direction, market positioning, business model, and long-term vision.

Volume I — Business & Functional Product Specification

Defines what RREP does from a business and functional perspective.

Volume II — Technical Architecture & Engineering Specification

Defines how RREP is technically designed and engineered.

Volume III — Implementation, Deployment & Operations Guide

Defines how RREP is implemented, deployed, maintained, monitored, upgraded, supported, and recovered.

Volume IV — UI/UX Design System & Product Standards

Defines how the product should look, behave, and maintain visual and interaction consistency.

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PART A — IMPLEMENTATION

CHAPTER 1 — INTRODUCTION

1.1 Purpose

The purpose of this guide is to establish a standardized process for deploying RREP as a standalone commercial platform for individual real estate organizations.

Every customer installation shall be treated as an independent deployment.

The implementation team shall avoid customer-specific modifications to the core source code whenever a configurable feature, module, workflow, setting, extension, or integration can satisfy the requirement.

1.2 Deployment Philosophy

RREP shall follow the principle:

Configure before customizing.

Customer-specific requirements should first be evaluated against:

1. Existing configuration
2. Existing permissions
3. Existing workflows
4. Existing modules
5. Existing feature flags
6. Existing integrations
7. Extension mechanisms

Only genuine product gaps should result in custom development.

1.3 Implementation Objectives

Every implementation shall achieve:

- Correct installation
- Secure configuration
- Correct licensing
- Correct module activation
- Correct organization setup
- Correct user setup
- Correct data configuration
- Successful backup configuration
- Successful monitoring
- Successful test transactions
- Successful customer acceptance

CHAPTER 2 — IMPLEMENTATION LIFECYCLE

A standard implementation shall follow these stages:

```
Customer Assessment
       ↓
Infrastructure Preparation
       ↓
License Provisioning
```

```
       ↓
Application Installation
       ↓
Database Initialization
       ↓
Configuration
       ↓
Module Activation
       ↓
Data Migration
       ↓
User Setup
       ↓
Testing
       ↓
Training
       ↓
Go-Live
       ↓
Support
```

2.1 Customer Assessment

Before installation, the implementation team shall document:

- Organization name
- Number of branches
- Number of users
- Number of properties
- Expected transaction volume
- Required modules
- Required integrations
- Hosting model
- Domain requirements
- Email requirements
- SMS requirements
- Payment requirements
- GIS requirements
- Backup requirements

2.2 Infrastructure Assessment

The technical team shall determine whether the customer requires:

- Cloud hosting
- VPS
- Dedicated server
- On-premises server
- Hybrid deployment

Infrastructure shall be sized according to expected workload.

CHAPTER 3 — ENVIRONMENT STRATEGY

RREP shall maintain separate environments where practical.

3.1 Development

Used by developers.

3.2 Testing

Used for:

- QA
- Integration testing
- User acceptance testing

3.3 Staging

Used to validate production-like deployments.

3.4 Production

Used by the customer.

Production credentials and data shall never be copied into development environments without appropriate controls and data protection procedures.

CHAPTER 4 — DEVELOPMENT ENVIRONMENT

Developers shall have access to the required development tooling.

Minimum requirements include:

- Git
- PHP
- Composer

- MySQL
- Node.js/npm where required
- Required PHP extensions
- Redis where required

Optional:

- Docker
- Local mail testing service
- Local object storage emulator
- Browser automation tools

4.1 Initial Setup

A new developer shall:

1. Clone the repository.
2. Install PHP dependencies.
3. Install frontend dependencies.
4. Configure environment variables.
5. Create a development database.
6. Run migrations.
7. Run seeders.
8. Build frontend assets.
9. Start required workers.
10. Run automated tests.

CHAPTER 5 — APPLICATION CONFIGURATION

RREP configuration shall be environment-driven.

Production configuration shall not be hard-coded into application source files.

Typical environment configuration includes:

```
APP_NAME
APP_ENV
APP_KEY
APP_URL

DB_CONNECTION
DB_HOST
DB_PORT
DB_DATABASE
```

```
DB_USERNAME
DB_PASSWORD

CACHE_STORE
QUEUE_CONNECTION

MAIL_MAILER
MAIL_HOST
MAIL_PORT
MAIL_USERNAME
MAIL_PASSWORD

FILESYSTEM_DISK

REDIS_HOST
REDIS_PORT
REDIS_PASSWORD
```

Additional variables shall be introduced for enabled integrations.

5.1 Environment Security

`.env` files shall:

- Never be committed to Git.
- Be readable only by authorized system users.
- Be backed up securely where required.
- Be protected from public web access.

CHAPTER 6 — DATABASE INITIALIZATION

The production database shall be created using the supported database version.

The implementation process shall include:

1. Create database.
2. Create restricted database user.
3. Assign required privileges.
4. Configure application credentials.
5. Run migrations.
6. Run required seeders.
7. Verify database integrity.

6.1 Production Database Principle

Database migrations shall be version-controlled.

Manual structural database changes shall be avoided unless documented and approved.

CHAPTER 7 — MODULE INSTALLATION

RREP shall use modular feature activation.

After base installation, the implementation team shall activate only the modules included in the customer's license.

Example:

```
Core
├── Property Management       ENABLED
├── CRM                       ENABLED
├── Sales                     ENABLED
├── Finance                   ENABLED
├── Survey                    DISABLED
├── Marketing                 ENABLED
├── AI                        DISABLED
└── Advanced BI               DISABLED
```

The interface shall automatically reflect module availability.

CHAPTER 8 — INITIAL SYSTEM CONFIGURATION

After installation, the administrator shall configure:

- Organization information
- Logo
- Address
- Contact information
- Currency
- Time zone
- Language
- Date format
- Number format
- Business identifiers

- Branches
- Departments
- Roles
- Permissions
- Document numbering
- Financial settings
- Notification settings

CHAPTER 9 — DATA MIGRATION

If the customer is migrating from another system, the implementation team shall perform a structured migration.

Possible source formats:

- Excel
- CSV
- Existing database
- API
- Legacy application exports

9.1 Migration Process

```
Source Assessment
       ↓
Data Mapping
       ↓
Data Cleaning
       ↓
Transformation
       ↓
Test Import
       ↓
Validation
       ↓
Customer Approval
       ↓
Production Import
```

9.2 Migration Principle

Never import unverified data directly into production.

A test migration shall be performed first.

CHAPTER 10 — CUSTOMER IMPLEMENTATION

Each customer shall receive an implementation plan.

The plan shall define:

- Scope
- Modules
- Users
- Timeline
- Responsibilities
- Data migration
- Integrations
- Training
- Acceptance criteria
- Go-live date

PART B — DEPLOYMENT

CHAPTER 11 — DEPLOYMENT ARCHITECTURE

A standard deployment shall consist of:

```
Internet
   │
   ▼
DNS
   │
   ▼
Nginx
   │
   ▼
Laravel Application
   │
 ┌─┴────────────────┐
 ▼                  ▼
MySQL              Redis
 │                  │
 ▼                  ▼
Data               Queue
                    │
                    ▼
                 Workers
```

Additional services may include:

- Object storage
- Search engine
- GIS
- AI
- Monitoring
- External messaging providers

CHAPTER 12 — SERVER REQUIREMENTS

Server requirements shall depend on customer scale.

Small Deployment

Suitable for small organizations:

- 2–4 CPU cores
- 4–8 GB RAM
- SSD storage
- MySQL
- Redis
- Linux

Medium Deployment

Recommended baseline:

- 4–8 CPU cores
- 8–16 GB RAM
- SSD/NVMe
- Dedicated database resources where appropriate

Enterprise Deployment

May require:

- Multiple application servers
- Dedicated database server
- Redis cluster
- Object storage
- Load balancing
- Monitoring infrastructure
- High-availability architecture

Actual sizing shall be determined by workload.

CHAPTER 13 — LINUX SERVER PREPARATION

Production servers shall use a supported Linux distribution.

The server shall be configured with:

- Firewall
- SSH hardening
- Automatic security updates where appropriate
- Time synchronization
- Monitoring
- Log rotation
- Backup agent where applicable

13.1 Firewall

Only required ports shall be exposed.

Typical public ports:

```
80
443
```

SSH access shall be restricted appropriately.

Database, Redis, and internal service ports should not be publicly exposed.

CHAPTER 14 — WEB SERVER CONFIGURATION

Nginx shall serve the Laravel application's public directory.

The web server shall:

- Route PHP requests through PHP-FPM.
- Serve static assets.
- Enforce HTTPS.

- Prevent access to protected files.
- Configure appropriate upload limits.
- Configure request timeouts.
- Configure security headers where appropriate.

The web root shall point to:

```
/public
```

The Laravel project root shall never be used directly as the public web root.

CHAPTER 15 — PHP CONFIGURATION

PHP shall be configured according to application requirements.

Relevant settings may include:

- Memory limit
- Maximum execution time
- Maximum upload size
- POST size
- OPcache
- Required extensions

OPcache shall be enabled in production.

CHAPTER 16 — MYSQL CONFIGURATION

MySQL shall be configured for:

- UTF-8-compatible character encoding
- Appropriate connection limits
- Appropriate buffer sizes
- Slow query monitoring
- Regular backups

Database credentials shall use a dedicated application user.

The application user shall receive only required permissions.

CHAPTER 17 — REDIS CONFIGURATION

Redis may be used for:

- Cache
- Queue
- Rate limiting
- Temporary application state

Redis shall not be publicly exposed.

Authentication and network restrictions shall be enabled where supported.

CHAPTER 18 — QUEUE WORKERS

Queue workers shall run continuously in production.

The recommended process manager shall be:

- Supervisor
- Systemd
- Container orchestration

Workers shall be monitored.

Failed jobs shall be reviewed regularly.

CHAPTER 19 — SCHEDULER

Laravel's scheduler shall run continuously through the system scheduler.

The scheduler shall execute tasks such as:

- Notifications
- Reminders
- Recurring reports
- Subscription/license checks
- Data cleanup
- Maintenance tasks
- Automated communications

Only one scheduler execution mechanism should be configured per server unless explicitly required.

CHAPTER 20 — STORAGE CONFIGURATION

The storage layer shall support:

- Property images
- Customer documents
- Contracts
- Receipts
- Survey files
- Reports
- Attachments

Storage shall enforce access control.

Private documents shall not be exposed through predictable public URLs.

CHAPTER 21 — SSL/TLS

Production systems shall use HTTPS.

Certificates may be provisioned through:

- Let's Encrypt
- Commercial certificate authorities
- Customer-managed certificates

Certificates shall be monitored for expiration.

HTTP requests should redirect to HTTPS.

CHAPTER 22 — DOMAIN CONFIGURATION

A customer deployment may use:

```
app.customer-domain.com
```

or:

```
realestate.customer-domain.com
```

DNS shall point the domain to the deployment server.

The implementation team shall verify:

- DNS resolution
- HTTPS
- Application URL
- Email links
- API URLs
- Webhooks

CHAPTER 23 — PRODUCTION DEPLOYMENT

A standard deployment process shall be:

1. Prepare server.
2. Configure DNS.
3. Install runtime.
4. Deploy source code.
5. Install Composer dependencies.
6. Build frontend assets.
7. Configure `.env`.
8. Generate application key where required.
9. Configure database.
10. Run migrations.
11. Configure storage.
12. Configure Redis.
13. Configure queue worker.
14. Configure scheduler.
15. Configure Nginx.
16. Configure SSL.
17. Configure monitoring.
18. Configure backups.
19. Activate license.
20. Run deployment tests.

CHAPTER 24 — DEPLOYMENT VERIFICATION

After deployment, verify:

Application

- Homepage loads.
- Login works.
- Dashboard works.
- Static assets load.

Database

- Connection works.
- Migrations completed.

Queue

- Jobs are processed.

Scheduler

- Scheduled jobs execute.

Storage

- Files upload.
- Files download according to authorization.

Email

- Test email succeeds.

SMS

- Test message succeeds if enabled.

APIs

- Authentication works.
- API endpoints respond correctly.

Security

- HTTPS works.
- Protected files are inaccessible.

- Debug mode is disabled.

PART C — LICENSING & ACTIVATION

CHAPTER 25 — PRODUCT LICENSING

RREP shall be commercially distributed as standalone licensed installations.

The customer purchases or subscribes to a defined product edition and feature set.

The license shall determine:

- Product edition
- Enabled modules
- Enabled features
- User limits where applicable
- Usage limits where applicable
- Expiration
- Support entitlement where applicable

CHAPTER 26 — LICENSE ACTIVATION

License activation shall associate an installation with an issued license.

Activation may use:

- License key
- Signed license file
- Activation token
- Secure activation API

The implementation shall avoid exposing proprietary license secrets.

CHAPTER 27 — FEATURE MODULE LICENSING

The licensing architecture shall support granular commercial packaging.

Example:

```
BASE
├── Property Management
├── CRM
└── Basic Reports

PROFESSIONAL
├── BASE
├── Sales
├── Finance
├── Marketing
└── Advanced Reports

ENTERPRISE
├── PROFESSIONAL
├── Survey
├── AI
├── Advanced BI
├── API
└── Advanced Integrations
```

The actual commercial packages may be changed without redesigning the application.

CHAPTER 28 — LICENSE VALIDATION

License validation shall verify:

- License authenticity
- Installation identity
- Expiry
- Enabled features
- Usage restrictions

Where online validation is used, temporary connectivity failure shall be handled according to the license policy rather than immediately disabling legitimate customer operations.

CHAPTER 29 — LICENSE RENEWAL

Renewal shall support:

- Extension of expiration
- Feature upgrades
- User limit changes
- Module activation
- Support renewal

License changes shall not require reinstalling the entire application unless technically necessary.

CHAPTER 30 — LICENSE FAILURE HANDLING

If a license becomes invalid:

1. Record the event.
2. Notify the administrator.
3. Preserve existing customer data.
4. Restrict only affected functionality according to policy.
5. Provide clear remediation information.

The platform shall not intentionally destroy or corrupt customer data because of licensing problems.

PART D — OPERATIONS

CHAPTER 31 — SYSTEM ADMINISTRATION

System administrators shall manage:

- Users
- Roles
- Permissions
- Branches
- Settings
- Modules
- Integrations
- Workflows
- Notifications
- Reports
- Audit logs

Administrative actions shall be audited.

CHAPTER 32 — USER ADMINISTRATION

Administrators shall be able to:

- Create users
- Disable users
- Assign roles
- Reset credentials
- Manage permissions
- Review activity
- Revoke sessions

Inactive employees shall have their accounts disabled promptly.

CHAPTER 33 — ORGANIZATION CONFIGURATION

Organization settings shall include:

- Legal name
- Trading name
- Logo
- Address
- Contact information
- Tax information where applicable
- Currency
- Time zone
- Default language
- Business identifiers

CHAPTER 34 — BRANCH CONFIGURATION

Branches shall have:

- Name
- Address
- Contact information
- Users
- Departments
- Properties
- Financial configuration where applicable

Branch access shall be controlled by permissions.

CHAPTER 35 — BUSINESS CONFIGURATION

Administrators shall configure:

- Property types
- Unit types
- Sales statuses
- Reservation statuses
- Payment methods
- Commission rules
- Document types
- Workflow rules
- Customer classifications
- Lead sources

CHAPTER 36 — NOTIFICATION ADMINISTRATION

Administrators shall configure:

- Email provider
- SMS provider
- WhatsApp provider
- Notification templates
- Sender identities
- Notification preferences

Delivery failures shall be visible in the administrative interface.

CHAPTER 37 — INTEGRATION ADMINISTRATION

Each integration shall have:

- Provider
- Credentials
- Status

- Configuration
- Connection test
- Error logs
- Activation/deactivation controls

Credentials shall be encrypted or securely stored.

CHAPTER 38 — DOCUMENT MANAGEMENT OPERATIONS

Administrators shall establish:

- Document categories
- Retention policies
- Access rules
- Required documents
- Versioning rules

Sensitive documents shall be restricted to authorized users.

CHAPTER 39 — FINANCIAL OPERATIONS

Financial administrators shall manage:

- Accounts
- Payment methods
- Transaction categories
- Invoices
- Receipts
- Payments
- Refunds
- Commissions
- Financial reports

Financial records shall maintain audit trails.

CHAPTER 40 — OPERATIONAL MAINTENANCE

Routine maintenance shall include:

- Reviewing logs
- Checking failed jobs
- Checking disk capacity
- Checking backups
- Checking SSL certificates
- Checking integrations
- Reviewing security alerts
- Reviewing system health

PART E — MONITORING & SUPPORT

CHAPTER 41 — MONITORING

Production monitoring shall cover:

- Server health
- Application health
- Database health
- Queue health
- Storage
- External integrations

CHAPTER 42 — LOGGING

Logs shall be centralized where practical.

Important categories include:

```
application
security
authentication
api
database
queue
payment
integration
ai
```

Logs shall have appropriate retention policies.

CHAPTER 43 — APPLICATION HEALTH

RREP shall expose a health-check mechanism.

Health checks should verify:

- Application availability
- Database connectivity
- Cache availability
- Queue availability
- Storage availability

Sensitive information shall never be returned by public health endpoints.

CHAPTER 44 — DATABASE MONITORING

The team shall monitor:

- Database size
- Connection count
- Slow queries
- Failed queries
- Lock contention
- Disk usage
- Backup status

CHAPTER 45 — QUEUE MONITORING

Queue monitoring shall track:

- Pending jobs
- Completed jobs
- Failed jobs
- Retry counts
- Processing time

Repeated failures shall trigger investigation.

CHAPTER 46 — STORAGE MONITORING

Storage monitoring shall track:

- Disk usage
- File count
- Growth rate
- Failed uploads
- Backup status

Alerts shall be configured before storage reaches critical capacity.

CHAPTER 47 — SECURITY MONITORING

Security monitoring shall cover:

- Failed login attempts
- Privilege changes
- New administrator accounts
- Suspicious API activity
- Unusual authentication activity
- Configuration changes
- Repeated authorization failures

CHAPTER 48 — INCIDENT MANAGEMENT

An incident is any event that significantly affects:

- Confidentiality
- Integrity
- Availability
- Financial operations
- Customer operations

- Security

48.1 Incident Levels

Critical

Complete outage, major security incident, severe data integrity problem.

High

Major module unavailable or major business operation affected.

Medium

Significant feature malfunction with workaround available.

Low

Minor defect or non-critical operational issue.

48.2 Incident Process

```
Detection
   ↓
Classification
   ↓
Containment
   ↓
Investigation
   ↓
Resolution
   ↓
Verification
   ↓
Documentation
   ↓
Root Cause Analysis
```

CHAPTER 49 — TROUBLESHOOTING

Application Not Loading

Check:

1. Nginx

2. PHP-FPM
3. Application logs
4. Environment configuration
5. Database connectivity
6. Disk space
7. Permissions

Login Failure

Check:

- User status
- Credentials
- Authentication logs
- Session configuration
- Database availability
- Rate limiting

Queue Failure

Check:

- Redis
- Worker status
- Failed jobs
- Worker logs
- Environment variables

Email Failure

Check:

- SMTP credentials
- DNS
- Provider status
- Application logs
- Queue status

File Upload Failure

Check:

- PHP upload limits
- Directory permissions
- Storage configuration
- Disk capacity
- File size restrictions

CHAPTER 50 — SUPPORT PROCEDURES

Support shall operate through a structured process.

Every support request should record:

- Customer
- Installation
- User
- Problem
- Severity
- Time reported
- Diagnostic information
- Resolution
- Root cause where applicable

Support personnel shall never request or store customer passwords.

PART F — BACKUP & RECOVERY

CHAPTER 51 — BACKUP STRATEGY

Backups shall protect:

- Database
- Uploaded files
- Configuration
- Critical application information

A recommended strategy is:

```
Daily Database Backup
+
```

```
Daily/Incremental File Backup
+
Periodic Full Backup
+
Offsite Backup
```

CHAPTER 52 — DATABASE BACKUP

Database backups shall be automated.

Backup procedures shall include:

- Timestamp
- Compression where appropriate
- Encryption where required
- Retention
- Offsite copy

CHAPTER 53 — FILE BACKUP

File storage shall be backed up independently of the database.

This is essential because database backups alone do not restore uploaded documents.

CHAPTER 54 — CONFIGURATION BACKUP

Critical configuration shall be recoverable.

This includes:

- Environment configuration
- Deployment configuration
- Nginx configuration
- Worker configuration
- Scheduler configuration
- License information
- Integration configuration

Secrets shall be backed up securely.

CHAPTER 55 — BACKUP VERIFICATION

A backup shall not be considered reliable merely because the backup job completed.

Restoration tests shall be performed periodically.

The team shall verify:

- Backup exists
- Backup is readable
- Database can be restored
- Files can be restored
- Application can reconnect
- Critical functionality works

CHAPTER 56 — RESTORE PROCEDURES

A standard restore procedure is:

```
Identify Incident
       ↓
Stop Affected Operations
       ↓
Preserve Current State
       ↓
Select Backup
       ↓
Restore Database
       ↓
Restore Files
       ↓
Restore Configuration
       ↓
Run Migrations if Required
       ↓
Verify Application
       ↓
Verify Data
       ↓
Resume Operations
```

CHAPTER 57 — DISASTER RECOVERY

Disaster recovery shall address:

- Server failure
- Disk failure
- Database corruption
- Accidental deletion
- Security incidents
- Hosting provider failure
- Natural disasters
- Major infrastructure outages

57.1 Recovery Objectives

Each customer deployment shall define:

RPO — Recovery Point Objective

Maximum acceptable amount of data loss.

RTO — Recovery Time Objective

Maximum acceptable recovery duration.

Enterprise customers may require stricter targets.

CHAPTER 58 — BUSINESS CONTINUITY

Where required, business continuity shall include:

- Secondary backups
- Alternative infrastructure
- Emergency contacts
- Recovery procedures
- Communication procedures
- Manual fallback procedures for critical business functions

PART G — UPDATES & RELEASES

CHAPTER 59 — RELEASE MANAGEMENT

Every release shall have:

- Version number
- Release date
- Changes
- Bug fixes
- Security fixes
- Database changes
- Configuration changes
- Migration instructions

CHAPTER 60 — APPLICATION UPDATES

A standard update shall be:

```
Backup
  ↓
Maintenance Mode
  ↓
Deploy New Code
  ↓
Install Dependencies
  ↓
Run Migrations
  ↓
Clear Caches
  ↓
Restart Workers
  ↓
Verify
  ↓
Disable Maintenance Mode
```

CHAPTER 61 — DATABASE MIGRATIONS

Database migrations shall:

- Be version-controlled.
- Be tested before production.

- Be backward-compatible where practical.
- Avoid destructive changes without migration planning.

Large migrations shall be planned carefully to minimize downtime.

CHAPTER 62 — CONFIGURATION CHANGES

Configuration changes shall be documented.

Production configuration should not be changed casually.

Critical changes shall be approved and recorded.

CHAPTER 63 — LICENSE CHANGES

License upgrades shall be designed to occur without unnecessary downtime.

Examples:

- Activating Finance
- Activating AI
- Increasing user limit
- Enabling API access
- Enabling advanced reporting

CHAPTER 64 — ROLLBACK

Every significant deployment shall have a rollback strategy.

Possible rollback mechanisms include:

- Previous application release
- Database backup restoration
- Reversible migrations
- Configuration restoration

Rollback procedures shall be tested periodically.

CHAPTER 65 — VERSION MANAGEMENT

RREP releases shall use a consistent versioning strategy.

Example:

```
Major.Minor.Patch

1.0.0
1.1.0
1.1.1
2.0.0
```

Major releases may contain breaking changes.

Minor releases should introduce compatible functionality.

Patch releases should primarily address bugs and security issues.

PART H — CUSTOMER HANDOVER

CHAPTER 66 — CUSTOMER ONBOARDING

Customer onboarding shall include:

- Organization setup
- User setup
- Module configuration
- Data import
- Integration configuration
- Workflow configuration
- Notification configuration

CHAPTER 67 — ADMINISTRATOR TRAINING

Administrators shall be trained on:

- User management
- Permissions
- Settings
- Modules
- Reports
- Backups
- Notifications
- Integrations
- Audit logs
- Troubleshooting basics

CHAPTER 68 — USER TRAINING

Training shall be role-specific.

Examples:

Sales Team

- Leads
- Customers
- Property listings
- Reservations
- Sales
- Follow-ups

Finance Team

- Invoices
- Payments
- Receipts
- Reconciliation
- Reports

Property Team

- Properties
- Units
- Occupancy
- Maintenance

Management

- Dashboards
- KPIs
- Reports

- Analytics

CHAPTER 69 — DOCUMENTATION HANDOVER

The customer shall receive appropriate documentation.

Depending on the agreement, this may include:

- User guide
- Administrator guide
- Integration documentation
- Backup instructions
- Support contacts
- License information
- Deployment information

Sensitive infrastructure credentials shall be transferred through secure mechanisms.

CHAPTER 70 — GO-LIVE CHECKLIST

Before go-live:

Infrastructure

- ☐ Server operational
- ☐ Domain configured
- ☐ HTTPS configured
- ☐ Firewall configured
- ☐ Monitoring active
- ☐ Backups active

Application

- ☐ Application deployed
- ☐ Database initialized
- ☐ Queue active
- ☐ Scheduler active

- [ ] Storage active

Business

- [ ] Organization configured
- [ ] Branches configured
- [ ] Users created
- [ ] Roles assigned
- [ ] Modules activated
- [ ] Workflows configured

Integrations

- [ ] Email tested
- [ ] SMS tested
- [ ] Payment integration tested
- [ ] GIS tested
- [ ] Other required integrations tested

Data

- [ ] Migration completed
- [ ] Data validated
- [ ] Customer approved

Security

- [ ] HTTPS active
- [ ] Debug disabled
- [ ] Credentials secured
- [ ] Admin access verified
- [ ] Backup verified

Acceptance

- [ ] Customer acceptance completed
- [ ] Training completed

- [ ] Go-live approved

CHAPTER 71 — POST-GO-LIVE SUPPORT

After go-live, the implementation team shall monitor the installation closely.

The first support period should focus on:

- User issues
- Data errors
- Configuration corrections
- Integration failures
- Performance
- Training gaps

Issues discovered during early operation shall be documented and classified as:

- Configuration issue
- User training issue
- Bug
- Enhancement
- Infrastructure issue
- Integration issue

PART I — OPERATIONAL STANDARDS

CHAPTER 72 — SECURITY OPERATIONS

Security shall be maintained continuously after deployment.

Operational security activities include:

- Reviewing access
- Disabling inactive users
- Updating dependencies
- Applying security patches
- Reviewing logs
- Monitoring suspicious activity

- Rotating credentials where necessary
- Reviewing administrator accounts

CHAPTER 73 — PERFORMANCE OPERATIONS

Performance shall be monitored continuously.

Indicators include:

- Average response time
- Peak response time
- Database query duration
- Queue processing time
- CPU usage
- Memory usage
- Disk usage
- API latency

Performance optimization shall be based on measured bottlenecks.

CHAPTER 74 — DATA GOVERNANCE

Customer data shall be treated as a critical business asset.

Operational policies shall define:

- Data ownership
- Data access
- Retention
- Export
- Archival
- Deletion
- Privacy
- Auditability

The platform shall not intentionally lock customer data into an inaccessible proprietary format.

CHAPTER 75 — MAINTENANCE WINDOWS

Planned maintenance should be communicated to customers in advance.

Maintenance activities may include:

- Application upgrades
- Database maintenance
- Security patches
- Infrastructure upgrades
- Backup testing
- Configuration changes

Critical security updates may require accelerated maintenance.

CHAPTER 76 — OPERATIONAL CHECKLISTS

DAILY CHECKLIST

- ☐ Application available
- ☐ Database healthy
- ☐ Queue healthy
- ☐ Failed jobs reviewed
- ☐ Backup completed
- ☐ Disk capacity acceptable
- ☐ Critical errors reviewed

WEEKLY CHECKLIST

- ☐ Review security logs
- ☐ Review failed integrations
- ☐ Review storage growth
- ☐ Review backup integrity
- ☐ Review system performance

- ☐ Review user/admin activity

MONTHLY CHECKLIST

- ☐ Test backup restoration
- ☐ Review user access
- ☐ Review inactive accounts
- ☐ Review infrastructure capacity
- ☐ Review dependencies
- ☐ Review security updates
- ☐ Review license status
- ☐ Review customer support issues

QUARTERLY CHECKLIST

- ☐ Disaster recovery exercise
- ☐ Security review
- ☐ Performance review
- ☐ Capacity planning
- ☐ Architecture review
- ☐ Backup strategy review
- ☐ License/product review

CHAPTER 77 — LONG-TERM PLATFORM OPERATIONS

RREP shall be operated as a continuously evolving commercial software product.

Long-term operational priorities shall include:

1. Security
2. Reliability
3. Performance
4. Customer satisfaction

5. Maintainability
6. Product evolution
7. Infrastructure efficiency
8. Data integrity

CUSTOMER INSTALLATION STANDARD

Every standalone RREP installation shall maintain a deployment record containing:

```
Customer
Installation ID
License
Version
Server
Domain
Database
Enabled Modules
Integrations
Administrator
Backup Configuration
Monitoring Configuration
Deployment Date
Last Update
Support Status
```

This record allows RehoSpace support personnel to understand the technical state of each customer deployment without requiring customer data to be stored centrally.

CUSTOMER-SPECIFIC CUSTOMIZATION POLICY

Customer-specific requirements shall be categorized as:

Configuration

Can be solved using existing platform settings.

Licensed Feature

Already exists but requires activation.

Extension

Can be implemented as a reusable extension.

Product Enhancement

Should be incorporated into the core RREP product.

Custom Development

A genuinely customer-specific requirement that should not become part of the standard product.

This classification is essential to maintaining RREP as a reusable commercial platform.

MULTI-CUSTOMER PRODUCT MANAGEMENT

Although RREP installations are standalone, RehoSpace shall maintain a central product engineering process.

RehoSpace shall manage:

- Source code
- Product releases
- Documentation
- License issuance
- Module packaging
- Security updates
- Product roadmap

Customer installations shall receive approved releases according to their support and licensing arrangements.

UPDATE DISTRIBUTION STRATEGY

Future versions may support controlled update channels:

```
Development
      ↓
Testing
      ↓
Stable Release
      ↓
Customer Deployment
```

Enterprise customers may optionally receive:

```
Early Access
Beta
Stable
Long-Term Support
```

SUPPORT BOUNDARIES

The commercial agreement shall clearly distinguish:

Included Support

Examples:

- Bug fixes
- Installation support
- Configuration support
- Security updates
- Product updates according to license

Billable Services

Examples:

- Custom features
- Data migration
- Custom integrations
- Major infrastructure changes
- Custom reports
- Customer-specific development
- On-site support

OPERATIONAL DATA ISOLATION

Each customer deployment shall maintain its own operational database.

Example:

```
Customer A
├── Application
├── Database
├── Files
└── Configuration

Customer B
├── Application
```

```
├── Database
├── Files
└── Configuration
```

Customer A must never be able to access Customer B's operational data through the application.

This architecture is a deliberate consequence of RREP's standalone commercial deployment model.

CENTRAL REHOSPACE SERVICES

Where required, RehoSpace may operate centralized services that do not contain customer operational data.

Examples:

- License activation
- License verification
- Update distribution
- Product telemetry where contractually permitted
- Documentation
- Support systems

Customer business data shall remain within the customer's deployment unless the customer explicitly authorizes a service requiring external data processing.

TELEMETRY PRINCIPLE

Any telemetry system shall be designed carefully.

The preferred approach is to collect technical information such as:

- Application version
- Module version
- Installation status
- Error category
- Performance metrics

without collecting unnecessary business information.

Telemetry shall respect contractual, privacy, and regulatory requirements.

END-TO-END DEPLOYMENT FLOW

The complete RREP deployment lifecycle shall be:

```
      CUSTOMER PURCHASE
              │
              ▼
     LICENSE PROVISIONING
              │
              ▼
  INFRASTRUCTURE ASSESSMENT
              │
              ▼
        SERVER SETUP
              │
              ▼
      RREP INSTALLATION
              │
              ▼
       DATABASE SETUP
              │
              ▼
   MODULE CONFIGURATION
              │
              ▼
      LICENSE ACTIVATION
              │
              ▼
        USER CREATION
              │
              ▼
        DATA MIGRATION
              │
              ▼
        INTEGRATIONS
              │
              ▼
           TESTING
              │
              ▼
          TRAINING
              │
              ▼
          GO-LIVE
              │
              ▼
         MONITORING
              │
              ▼
          SUPPORT
              │
              ▼
   UPDATES & MAINTENANCE
```

FINAL OPERATIONAL PRINCIPLE

RREP shall be deployed and operated as a **repeatable commercial product**.

The objective is not to create a different technical system for every real estate customer.

Instead:

```
                   RREP PRODUCT
                        │
         ┌──────────────┼────────────┐
         │              │            │
    Customer A   Customer B   Customer C
         │              │            │
    Standalone   Standalone   Standalone
    Deployment  Deployment   Deployment
         │              │            │
    Configuration Configuration Configuration
    + Modules     + Modules    + Modules
```

The core product remains standardized.

Customer differences should primarily be represented through:

- Licensing
- Module selection
- Configuration
- Permissions
- Workflows
- Integrations
- Branding
- Optional extensions

This ensures that RehoSpace can sell RREP repeatedly without creating a separate software product for every customer.

TECHNICAL OPERATIONS PRINCIPLE

Every production installation must satisfy the following:

Secure

The installation is protected against unauthorized access.

Reliable

The application and supporting services operate consistently.

Recoverable

Backups exist and restoration is possible.

Observable

The technical team can determine whether the system is healthy.

Maintainable

The installation can be updated without unnecessary complexity.

Supportable

The technical team can diagnose and resolve problems.

Reusable

The deployment process can be repeated for another customer.

FINAL DEPLOYMENT ACCEPTANCE CRITERIA

An RREP installation shall not be considered production-ready until:

- Application is successfully deployed.
- Database is operational.
- Required modules are activated.
- License is valid.
- Users are configured.
- Permissions are verified.
- Required integrations are tested.
- HTTPS is active.
- Production debug mode is disabled.
- Queue workers are operational.
- Scheduler is operational.
- File storage is operational.
- Backups are operational.
- Restore capability has been verified.
- Monitoring is operational.
- Critical workflows have been tested.
- Customer acceptance has been obtained.
- Documentation has been handed over.

FINAL STATEMENT

The purpose of Volume III is to establish a repeatable operational discipline around RREP.

A successful enterprise product is not defined only by its source code. It is defined by the ability to deploy that software consistently, protect its data, monitor its health, recover from failures, update it safely, support customers effectively, and evolve it without destabilizing existing installations.

RREP shall therefore be operated according to a product-centered engineering model in which:

One reusable product can serve many customers through independent, configurable, licensed deployments.

This principle protects the long-term maintainability of the platform while allowing RehoSpace to scale its commercial reach across the real estate market.

END OF VOLUME III

RehoSpace Real Estate Platform (RREP)

Implementation, Deployment & Operations Guide

Version 1.0
